

Quantum Mechanics III

Homework 9

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Topic

Correlation Experiments

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Correlation Experiments. State all assumptions and conventions.

$$E(a, b) = \sum_{x, y = \pm 1} xy p(x, y | a, b)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b')$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$|S| \leq 2 \text{ for registered factorized models}$$

Show enough intermediate work that another student can reproduce the calculation.